
A Fragile Number Sense: Probing the Elemental Limits of Numerical Reasoning in LLMs

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Abstract

Large Language Models (LLMs) have demonstrated remarkable emergent capabilities, yet the robustness of their numerical reasoning remains an open question. While standard benchmarks evaluate LLM reasoning on complex problem sets using aggregated metrics, they often obscure foundational weaknesses. In this work, we probe LLM mathematical numeracy by evaluating performance on problems of escalating complexity, from constituent operations to combinatorial puzzles. We test several state-of-the-art LLM-based agents on a 100-problem challenge comprising four categories: (1) basic arithmetic, (2) advanced operations, (3) primality checking, and (4) the Game of 24 number puzzle. Our results show that while the agents achieved high accuracy on the first three categories, which require deterministic algorithmic execution, they consistently failed at the number puzzle, underlining its demand for a heuristic search over a large combinatorial space to be a significant bottleneck. These findings reveal that the agents' proficiency is largely confined to recalling and executing known algorithms, rather than performing generative problem-solving. This suggests their apparent numerical reasoning is more akin to sophisticated pattern-matching than flexible, analytical thought, limiting their potential for tasks that require novel or creative numerical insights.

Keywords: LLMs; LRMs; Numerical Reasoning; Problem Solving

1 Introduction

LLMs have demonstrated exceptional abilities at diverse tasks, such as writing code, composing essays, and solving mathematical problems. A recent evolution is specifically designed for tasks involving reasoning, referred to as Large Reasoning Models (LRMs). However, most of their perceived reasoning abilities are examined using final performance on large benchmark problem sets, while the actual reasoning processes remain less explored.

LLMs (and LRMs) reason through chains and trees of thoughts, guided by reinforcement learning from sources like human feedback or simulation-based optimization. To understand the reasoning processes, their constituent elements need to be investigated. Importantly, at each step of the chains or the trees, the models reason by probabilistic pattern matching and generate responses as probabilistic token predictions, each with less than 100% probability of success. Therefore, as the number of reasoning steps increases, the probability of accuracy drops exponentially. Moreover, if the LLMs are weak at one or a few specific steps, the weakness may propagate and cause the trace to diverge towards inaccurate solutions. Evaluations relying on large aggregate benchmarks would obscure this fragility of reasoning that may compromise the model's reliability and safety in complex environments.

We explore the reasoning abilities of LLMs in the domain of mathematical reasoning and problem solving. A key point to consider is the difference between natural language understanding and

understanding mathematical concepts. LLMs are designed to capture patterns in a finite space of words, but the grammar of numbers and the mathematical operations thereon is essentially infinite – a universe humans explore by exploiting the numerous characteristics of numbers and relationships among them. To illustrate, we consider prime numbers, prime factorizations, transcendental numbers, or the errors introduced when representing them through decimal fractions. Our understanding of these concepts allows us to solve problems of enormous complexity using limited computational resources. We view this human ability to solve problems much more efficiently than permitted by brute-force methods as the core of reasoning abilities – a view that captures the common theme that runs through several established frameworks of human rationality: bounded rationality (Simon, 1955), ecological rationality (Gigerenzer, 2001), and resource rationality (Lieder and Griffiths, 2020).

As it is difficult to be efficient in solving mathematical problems purely based on pattern matching and without understanding the concepts, we choose mathematical reasoning as our experimental paradigm. We examine the mathematical reasoning abilities of LLM agents based on their ability to identify the elements of problems (i.e., numbers, operations, and their relationships) and use them to solve complex problems efficiently. We begin our investigation from low levels (e.g., performing operations and calculating expressions) before gradually moving towards higher levels, with a goal to identify where the cracks in reasoning appear and affect complex problem solving performance. As we demonstrate, the LLMs perform reasonably well in the elementary tasks, but they struggle to use these elements in efficiently solving complex problems with uncertainty that require trial-and-error search. Thereafter, we replicate these findings for more advanced LRMs, showing the struggles with efficient search to be a common theme.

2 Relevant Research on Reasoning

2.1 Evaluating LLM Reasoning

Reasoning entails making a series of rational decisions using available information and computational resources to achieve specific goals. Mathematical reasoning involves solving math problems by efficiently using mathematical concepts and their relationships. LLM reasoning is typically evaluated via aggregate scores on large benchmarks. For example, the MATH dataset (Hendrycks et al., 2021) contains over 12500 problems, while the Minif2f dataset (Zheng et al., 2021) includes 488 problems from prestigious math competitions (e.g., AIME, AMC, and IMO). Reasoning has also been studied as a component of general intelligence, as in the Abstract Reasoning Corpus for Artificial General Intelligence (ARC AGI) benchmark (Chollet, 2019). While these benchmarks proved challenging for then state-of-the-art agents, LLMs have improved rapidly since (Garisto, 2022; Chollet et al., 2024). However, aggregating performance over a large number of problems obscures the sources of improvement, leading researchers to seek more granular explanations of LLM reasoning.

2.2 Explaining LLM Reasoning

Wei et al. (2022) showed that *Chain of Thought* (CoT) prompting – simply asking models to divide complex problems into smaller subproblems and reason step by step – considerably improves reasoning performance without additional training. The self-reported steps serve as explanations of reasoning. Extending CoT, Yao et al. (2023) introduced the *Tree of Thought* (ToT) approach that represents problems as trees of subproblems before systematically searching through numerous reasoning paths. Recent efforts integrate tree-search methods with reinforcement learning from human feedback (e.g., Bai et al., 2022) or self-reflections (e.g., Ouyang et al., 2022). CoT, ToT, and reinforcement learning have become integral elements of modern LLM agents’ reasoning abilities.

Despite their usefulness, the LLM self-reports have been questioned as to whether they reflect the true processes used in reaching solutions (Ahn et al., 2024; Turpin et al., 2023). To specify the details of the reasoning processes, Explainable AI (XAI) offers several classes of tools of varying granularity (Bereska and Gavves, 2024). A particularly promising approach has been mechanistic interpretability (MI), which seeks to explain behavior by reverse-engineering MLP-based AI models from neuron-level information. For example, Bricken et al. (2023) identified interpretable features corresponding to distinct reasoning operations from neural activations. Nanda et al. (2023) reverse-engineered the algorithm learned by a single-layer transformer for modular addition (i.e., problems of the form $\text{mod}(a + b, n)$). While MI has been useful in deciphering LLM reasoning, scaling for more complex problems remains a challenge (Bereska and Gavves, 2024).

Shojaee et al. (2025) examined how problem complexity affects reasoning by designing controllable puzzle environments – such as the Tower of Hanoi (ToH) and river crossing tasks – that allow complexity to vary while keeping the environment constant. Their findings reveal that although LRMs appear to “think” better, their reasoning remains fragile. On low-complexity problems, standard LLMs outperformed LRMs; on medium-complexity tasks, LRMs had the edge. Yet both failed on high-complexity problems, underscoring the limits of LLM reasoning. The collapse in ToH performance is particularly surprising, given the existence of deterministic optimal strategies that apply across all complexity levels. Humans tend to discover these strategies rapidly through learning (see Section 2.3). Moreover, Shojaee et al. observed that the LRMs sometimes overlooked the right solutions and continued towards worse solutions, suggesting a lack of basic problem understanding. A primary goal of our work is to assess the LLMs’ understanding of mathematical elements as a step toward comprehending their overall problem-solving processes.

2.3 Explaining Human Reasoning

The ToH task, with its vast search space, challenges new learners but also contains discoverable subgoals that can simplify the problem. Anzai and Simon (1979) investigated how an individual unfamiliar with the ToH task discovered new strategies as they learned to solve a five-disk version over four trials. Their strategy evolved in each attempt – from selective search, to goal-peg planning using means-end analysis, then to a recursive subgoal strategy, and finally to a pyramid strategy that treated several disks as sets. Notably, each of the last three strategies yields optimal performance, but the latter strategies become more efficient to represent and execute. Previously, Simon and Reed (1976) examined the strategy shifts in a variant of the river crossing problem. They showed that in such complex problems, an “all-or-none” approach in search of optimality is impossible; rather, we need to use heuristics to find acceptable solutions that satisfy and suffice for our goals within limited resources. This principle forms the basis of the bounded rationality in humans that enables us to solve complex problems using limited computational resources (Simon, 1955; Newell et al., 1958; Simon and Newell, 1971; Gigerenzer, 2001, 2020).

Specific to mathematical reasoning, we possess an abstract number sense – an intuitive grasp of numbers, their relationships, and manipulations – that enables us to quickly estimate and reason without formal calculations (Dehaene, 2001). Its origins have been explored from multiple perspectives. Gallistel and Gelman (1992) described two systems: a preverbal one for approximating quantities early in development and a verbal one for precise calculations shaped by cultural learning. Dehaene (1992) proposed a triple code model: visual numerals (e.g., “37”), verbal labels (“thirty seven”), and analog magnitudes distributed along a number line. While number sense is abstract, neuroimaging studies reveal a specific brain region (the inferior parietal cortex) that contributes heavily to it (Dehaene, 1996; Kiefer and Dehaene, 1997; Pinel et al., 1999). A part of number sense is biologically determined and has evolutionary roots, and a part is learned and developed.

Generally, studies of human reasoning and developmental changes sought boundedly rational explanations through processes or strategies that are implementable within available resources, as opposed to a black box relating input information to output performance. For example, while learning addition, children have been observed to adaptively use at least five different strategies (Siegler and Shrager, 1984; Siegler, 1987) and their strategy transitions with learning have been emulated in cognitive models (Shrager and Siegler, 1998). For the ToH, Anzai and Simon (1979) predicted the sequence in which a learner would discover improved strategies. For an accurate understanding of LLM reasoning, especially in mathematical problems, such process-level explanations are essential.

3 Objectives, Framework, & Application

Do LLMs acquire a true *number sense*? Human numerical intuition allows us to simplify complex problems and navigate vast solution spaces efficiently. Without this sense, LLMs may excel at executing deterministic calculations but are likely to falter when tasks demand heuristic search, such as in trial-and-error problem-solving. This makes mathematical challenges an ideal paradigm for investigating how LLMs integrate elementary skills into complex strategies. While some mechanistic studies show that LLMs encode basic mathematical relationships, how these are integrated for higher-order reasoning remains less explored. This is partly because existing benchmarks and interpretability